

API SECURITY RISK ANALYSIS

Task Title:

API Security Risk Analysis

Submitted By

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1. Introduction

Application Programming Interfaces (APIs) are widely used to enable communication between different software applications. Because APIs often expose data and functionality, they can become targets for security attacks if not properly protected.

This report analyzes a public API using Postman and identifies potential security risks, their impact, and recommended mitigation measures.

2. Objective

The objective of this task is to:

- Understand how APIs work.
 - Test API endpoints using Postman.
 - Identify common security risks.
 - Evaluate the impact of these risks.
 - Recommend security improvements.
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3. Tools Used

- Postman
 - Public Test API (JSONPlaceholder)
 - Web Browser
 - GitHub
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4. API Tested

Base URL

<https://jsonplaceholder.typicode.com>

Endpoints Tested

- GET /users
 - GET /posts
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5. Testing Methodology

The API endpoints were accessed using the GET method in Postman. The responses were examined to identify security weaknesses, exposed information, authentication requirements, and other security-related observations.

Screenshots of the responses were captured as evidence of testing.

6. Findings

Finding 1: No Authentication Mechanism

Severity: Medium

Observation:

The API endpoints can be accessed without requiring authentication credentials.

Risk:

Unauthorized users may access available information without verification.

Recommendation:

Implement authentication mechanisms such as API Keys, OAuth 2.0, or JWT tokens.

Finding 2: Excessive Data Exposure

Severity: Medium

Observation:

The API response contains user-related information such as names, email addresses, and address details.

Risk:

Sensitive information may be exposed to unintended users.

Recommendation:

Return only the necessary fields required by the application.

Finding 3: Missing Rate Limiting

Severity: Low

Observation:

No visible rate-limiting controls were identified during testing.

Risk:

Attackers could send a large number of requests, potentially causing service degradation.

Recommendation:

Implement request throttling and rate-limiting mechanisms.

7. Risk Assessment Summary

Finding	Severity
No Authentication	Medium
Excessive Data Exposure	Medium
Missing Rate Limiting	Low

8. Recommendations

1. Implement strong authentication controls.
2. Limit exposed information in API responses.
3. Apply rate limiting to prevent abuse.
4. Monitor API traffic and logs.
5. Perform regular security assessments.

9. Conclusion

The API Security Risk Analysis demonstrated several common API security concerns, including lack of authentication, excessive data exposure, and absence of visible rate-limiting controls. While the tested API is intended for educational purposes, the identified issues highlight the importance of secure API design and implementation in real-world applications.

